ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

# DATA COMMUNICATION AND NETWORKING

## PAPER 1 · PRACTICE A

**MARKS: 100** **TIME: 3 hours**

**NC(V) Information Technology and Computer Science, Level 4. Section A: objective questions (30) · Section B: network models and protocols, IP addressing and subnetting, devices, media and topologies, wireless networks, network security and troubleshooting (70).**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

## INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL calculations.
5. Write neatly and legibly.

## SECTION A

## QUESTION 1

**1.1** Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 Which layer of the OSI model is responsible for routing packets between networks?

A Physical
B Data link
C Network
D Session

1.1.2 A MAC address is …

A a 48-bit physical address burned into a network card.
B a 32-bit logical address.
C a domain name.
D a port number.

1.1.3 Which protocol automatically assigns IP addresses to devices on a network?

A DNS
B FTP
C SMTP
D DHCP

1.1.4 The default subnet mask of a Class C network is …

A 255.0.0.0.
B 255.255.255.0.
C 255.255.0.0.
D 255.255.255.255.

1.1.5 Which device forwards frames only to the port where the destination MAC address is connected?

A Hub
B Repeater
C Modem
D Switch

1.1.6 HTTPS normally uses port …

A 21.
B 443.
C 25.
D 80.

1.1.7 Which cable is immune to electromagnetic interference and suitable for long distances?

A UTP Cat 5e
B Coaxial
C Fibre-optic
D STP

1.1.8 Which of the following is a private IP address?

A 192.168.1.20
B 8.8.8.8
C 41.0.12.5
D 196.25.1.1

1.1.9 The protocol that translates domain names into IP addresses is …

A ARP.
B DNS.
C ICMP.
D NAT.

1.1.10 The command used to test whether a host can be reached is …

A ipconfig.
B format.
C chkdsk.
D ping.

(10)

**1.2** Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 TCP is connection-oriented and guarantees delivery of data.

1.2.2 UDP is preferred for live video calls because it has less overhead.

1.2.3 In a star topology, all devices connect to a central switch.

1.2.4 An IPv6 address is 32 bits long.

1.2.5 WPA3 is a more secure Wi-Fi standard than WEP.

1.2.6 A router connects different networks.

1.2.7 The maximum recommended length of a UTP Ethernet cable run is 500 metres.

1.2.8 Bandwidth is the amount of data that can be transmitted per second.

1.2.9 A VPN creates an encrypted connection over a public network.

1.2.10 A crossover cable is the standard cable for connecting a computer to a modern switch.

(10)

**1.3** Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

| | **COLUMN A** | | **COLUMN B** |
|---|---|---|---|
| 1.3.1 | Translates private IP addresses to one public address | A | denial of service (DoS) |
| 1.3.2 | Finds the MAC address that belongs to an IP address | B | MAN |
| 1.3.3 | Used to send e-mail between mail servers | C | RJ45 |
| 1.3.4 | Network that covers a city or campus | D | latency |
| 1.3.5 | Logical division of a switch into separate broadcast domains | E | firewall |
| 1.3.6 | Connector used on twisted-pair Ethernet cables | F | ARP |
| 1.3.7 | Name that identifies a wireless network | G | SMTP |
| 1.3.8 | Device or software that filters traffic according to rules | H | NAT |
| 1.3.9 | Time it takes for data to travel from source to destination | I | SSID |
| 1.3.10 | Attack that floods a server with traffic so it cannot respond | J | VLAN |

(10)

**TOTAL SECTION A: [30]**

## SECTION B

## QUESTION 2: NETWORK MODELS AND PROTOCOLS

**2.1** Name the SEVEN layers of the OSI model, from layer 7 to layer 1. (7)

**2.2** Match the four layers of the TCP/IP model to the OSI layers. (4)

**2.3** Compare TCP and UDP under the headings: connection, reliability and an example of use. (6)

**2.4** Explain the purpose of each of the following protocols: (a) FTP; (b) DHCP; (c) IMAP. (3)

**[20]**

## QUESTION 3: IP ADDRESSING AND SUBNETTING

**3.1** Convert the IP address 172.16.45.130 to binary dotted notation. (4)

**3.2** A company has the network address 192.168.10.0/24 and needs four subnets of equal size.

3.2.1 How many bits must be borrowed from the host part? (1)

3.2.2 Give the new subnet mask in dotted-decimal and in CIDR notation. (2)

3.2.3 How many usable host addresses does each subnet have? Show your calculation. (2)

3.2.4 Give the network address, the first usable host, the last usable host and the broadcast address of the SECOND subnet. (4)

**3.3** Give TWO advantages of IPv6 over IPv4. (2)

**[15]**

## QUESTION 4: DEVICES, MEDIA AND TOPOLOGIES

> A small accounting firm on two floors has 25 computers, 3 network printers, a file server and Wi-Fi for visitors. It connects to the Internet through a fibre line.

**4.1** Name FIVE network devices the firm needs and state the function of EACH. (10)

**4.2** Recommend a topology for the firm and give TWO reasons. (3)

**4.3** Explain how the two floors could be connected if the distance between the switches is 120 m. (2)

**4.4** Explain why the visitors' Wi-Fi should be on a separate network from the staff computers. (2)

**4.5** Explain the difference between a peer-to-peer and a client–server network, and say which the firm should use. (3)

**[20]**

## QUESTION 5: SECURITY AND TROUBLESHOOTING

**5.1** Give FIVE measures to secure the firm's network. (5)

**5.2** A staff member cannot access the Internet. Running ipconfig shows the IP address 169.254.12.7. Explain what this means and give TWO steps to fix the problem. (4)

**5.3** Explain what the tracert (traceroute) command shows. (2)

**5.4** Explain the difference between symmetric and asymmetric encryption. (4)

**[15]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF QUESTION PAPER**